

4.1

$$\begin{array}{l|l} A (0,0,1) & 1 = c \\ B (1,0,1) & 1 = a + c \\ C (0,1,1) & 1 = b + c \\ D (1,1,2) & 2 = a + b + c \\ E (1,-1,0) & 0 = a - b + c \\ F (-1,1,1) & 1 = -a + b + c \end{array}$$

$$\begin{array}{ccc|c} a & b & c & \\ \hline 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{array}$$

||
A ||
→
b

rovina:

Řešení: $\hat{x} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad (x_i, y_i, z_i) \in \mathbb{R}^3, \quad z = ax + by + c$

Metoda nejmenších čtverců

s $\sum (z_i - (ax_i + by_i + c))^2$... co nejmenší

$$A^T A \hat{x} = A^T \vec{b}$$

Výpočet:

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}_{3 \times 6} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix}_{6 \times 3} = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} \leadsto \text{SLR: } \begin{pmatrix} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{pmatrix}$$

$$\sim \begin{pmatrix} -1 & 4 & 2 & 4 \\ 4 & -1 & 2 & 2 \\ 2 & 2 & 6 & 6 \end{pmatrix} \sim \begin{pmatrix} -1 & 4 & 2 & 4 \\ 0 & 15 & 10 & 18 \\ 0 & 10 & 10 & 14 \end{pmatrix} \sim \begin{pmatrix} -1 & 4 & 2 & 4 \\ 0 & 5 & 0 & 4 \\ 0 & 5 & 5 & 7 \end{pmatrix} \begin{cases} b = 4/5 \\ c = 3/5 \\ a = 2/5 \end{cases}$$

Řešení:

$$\hat{x} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2/5 \\ 4/5 \\ 3/5 \end{pmatrix}, \quad \boxed{z = 0,4x + 0,8y + 0,6}$$

III $\sum (z_i - (ax_i + by_i + c))^2 = 0,22$ [... geometrie]